



MINI WIND TURBINE FOR URBAN ENERGY GENERATION



A PROJECT REPORT

Submitted by

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In partial fulfillment for the award of the degree

of

BACHELOR OF ENGINEERING

IN

MECHANICAL ENGINEERING

**KNOWLEDGE INSTITUTE OF TECHNOLOGY, SALEM
(An Autonomous Institution)**

ANNA UNIVERSITY : CHENNAI 600 025

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BONAFIDE CERTIFICATE

Certified that this project report “**MINI WIND TURBINE FOR URBAN ENERGY GENERATION**” is the bonafide work of “**MANIKANDAN K, MOHAN G, NAVEENKUMAR R, SABARI K**” who carried out the project work under my supervision.

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ABSTRACT

The increasing demand for electrical energy and the growing concern about environmental pollution have created a need for sustainable and renewable energy sources. Wind energy is one of the most promising renewable resources due to its clean, eco-friendly, and cost-effective nature. This project presents the design and development of a Mini Wind Turbine for Urban Energy Generation capable of converting low-speed urban wind into useful electrical energy. The system uses PVC turbine blades and a DC generator to generate electricity, which is stored in a rechargeable 18650 lithium-ion battery through a TP4056 charging module. An ESP32 microcontroller is employed to monitor important parameters such as wind speed, voltage, current, and power generation. The measured data is displayed on an OLED display and transmitted to the Blynk IoT platform for real-time remote monitoring. The integration of IoT technology enables efficient energy management and performance analysis from any location. The proposed system provides a compact, low-cost, and environmentally friendly solution for urban renewable energy generation. It can be implemented in residential buildings, educational institutions, and smart city applications. The project promotes the effective utilization of renewable energy resources and contributes to sustainable urban development.

Keywords: Wind Energy, Mini Wind Turbine, Urban Energy Generation, Renewable Energy, ESP32, Internet of Things (IoT), Blynk, Wind Speed Sensor, Power Monitoring, Sustainable Energy.

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LIST OF ABBREVIATIONS

ABBREVIATIONS	MEANING
ESP32	Espressif32-bit Microcontroller
IoT	Internet of Things
OLED	Organic Light Emitting Diode
PWM	Pulse Width Modulation
DC	Direct Current
AC	Alternating Current
V	Voltage
A	Current
W	Watt

CHAPTER 1

INTRODUCTION

The increasing demand for electricity due to rapid urbanization and population growth has created a significant need for sustainable and renewable energy sources. Conventional energy resources such as coal, oil, and natural gas are becoming depleted and contribute to environmental problems including air pollution and global warming. As a result, renewable energy technologies have gained considerable attention worldwide. Among these technologies, wind energy is recognized as one of the most efficient and environmentally friendly sources of power generation. Wind energy converts the kinetic energy of moving air into electrical energy without producing harmful emissions, making it an ideal solution for sustainable development.

A Mini Wind Turbine is a compact renewable energy system designed to generate electricity from low and moderate wind speeds commonly found in urban environments. Unlike large wind turbines that require vast open spaces, mini wind turbines can be installed on rooftops, residential buildings, educational institutions, and commercial structures. These systems utilize available urban wind resources to produce clean energy for small-scale applications. The generated electrical energy can be stored in rechargeable batteries and used to power electronic devices, lighting systems, and monitoring equipment, thereby reducing dependence on conventional grid electricity.

This project focuses on the development of a Mini Wind Turbine for Urban Energy Generation integrated with an ESP32-based monitoring system. The turbine converts wind energy into electrical energy, which is stored in a battery through a charging circuit. Sensors are used to measure wind speed, voltage, and current, while the ESP32 processes and displays the collected data on an OLED display and

transmits it to the Blynk IoT platform for remote monitoring. The proposed system provides a cost-effective, eco-friendly, and smart energy solution that promotes renewable energy utilization in urban areas and supports the development of sustainable and energy-efficient cities.

1.1 BACKGROUND OF RENEWABLE

Renewable energy is energy generated from naturally replenishing resources such as sunlight, wind, water, biomass, and geothermal heat. Unlike fossil fuels, which are limited and contribute to environmental pollution, renewable energy sources are sustainable, clean, and environmentally friendly. The increasing global demand for electricity, depletion of conventional energy resources, and growing concerns about climate change have encouraged governments and industries worldwide to invest in renewable energy technologies. These energy sources help reduce greenhouse gas emissions, improve energy security, and support sustainable economic development.

Among the various renewable energy sources, wind energy has become one of the fastest-growing and most widely adopted technologies due to its reliability and availability. Wind turbines convert the kinetic energy of moving air into electrical energy without producing harmful emissions. While large wind farms are commonly used for large-scale power generation, small and mini wind turbines provide an effective solution for urban and residential applications. These systems enable clean energy generation at the local level, reduce dependence on conventional electricity, and contribute to the development of sustainable and smart energy systems for future generations.



FIG 1.1 WIND ENERGY CONCEPT

1.2 WIND ENERGY IN URBAN AREAS

Wind energy is increasingly being recognized as a viable renewable energy source for urban environments due to the growing demand for clean and sustainable electricity. Urban areas experience wind flow around rooftops, high-rise buildings, bridges, and open spaces, which can be effectively utilized using mini wind turbines designed for low and moderate wind speeds. These compact systems can be installed on residential, commercial, and institutional buildings to generate electricity locally, reducing dependence on conventional power sources and minimizing transmission losses. Urban wind energy systems are environmentally friendly, require relatively low maintenance, and can be integrated with smart monitoring technologies to optimize performance. By harnessing available wind resources within cities, mini wind turbines contribute to reducing carbon emissions, promoting energy sustainability, and supporting of smart and eco-friendly urban communities.

1.3 NEED FOR SUSTAINABLE POWER GENERATION

Mini wind turbines have emerged as an effective solution for decentralized energy generation in urban and semi-urban areas. With increasing electricity demand and rising energy costs, there is a need for alternative energy sources that are environmentally friendly and economically viable. Mini wind turbines can utilize available wind resources to generate electricity on a small scale without requiring large installations.

These systems are particularly suitable for residential buildings, educational institutions, and commercial establishments where space is limited. They help reduce dependence on conventional grid electricity and contribute to sustainable energy development. Their compact design, low maintenance requirements, and ease of installation make them an attractive renewable energy solution.

1.4 CHALLENGES IN URBAN ENERGY SYSTEMS

Urban energy systems face several challenges due to the rapid growth of population, industrialization, and increasing electricity demand. Cities consume a large amount of energy for residential, commercial, transportation, and industrial activities, placing significant pressure on existing power infrastructure. This often leads to energy shortages, peak load problems, and increased dependence on conventional fossil-fuel-based power generation. As urban populations continue to expand, ensuring a reliable and sustainable energy supply becomes increasingly difficult.

Another major challenge is the environmental impact associated with traditional energy production. The excessive use of fossil fuels contributes to air pollution, greenhouse gas emissions, and climate change, which directly affect urban living conditions. Additionally, limited space for large-scale renewable energy

installations, high energy transmission losses, and the rising cost of electricity further complicate urban energy management. These challenges highlight the need for decentralized renewable energy solutions, such as mini wind turbines, that can generate clean energy locally and support sustainable urban development.

1.5 MINI WIND TURBINE TECHNOLOGY

Mini wind turbine technology is designed to generate electrical energy from low and moderate wind speeds using compact wind turbines. These systems are smaller than conventional wind turbines and are suitable for urban and residential applications.

A mini wind turbine consists of blades, a generator, a charging circuit, and an energy storage system. The rotating blades convert the kinetic energy of wind into mechanical energy, which is then converted into electrical energy by the generator.

Modern mini wind turbines can be integrated with sensors, microcontrollers, and IoT platforms for real-time monitoring. This technology provides a clean, renewable, and cost-effective solution for decentralized power generation in urban environments.

1.6 SCOPE OF THE PROJECT

This project focuses on the design and development of a mini wind turbine system for generating electricity in urban environments. The system utilizes available wind energy and converts it into usable electrical power for small-scale applications.

The project also includes real-time monitoring of wind speed, voltage, current, and power using an ESP32 microcontroller and IoT technology. It can be implemented in homes, educational institutions, and commercial buildings to promote sustainable energy generation.

1.7 OBJECTIVES OF THE PROJECT

The primary objective of this project is to design and develop a mini wind turbine system capable of generating electrical energy from available urban wind resources. The system aims to provide a clean and sustainable alternative to conventional power sources.

Another objective is to monitor wind speed, voltage, current, and power generation using an ESP32 microcontroller and IoT technology. The collected data is displayed on an OLED screen and transmitted to the Blynk platform for real-time remote monitoring and analysis.

CHAPTER 2

LITERATURE REVIEW

2.1 URBAN RENEWABLE ENERGY SYSTEMS

Urban renewable energy systems are designed to generate clean and sustainable energy within cities using resources such as wind, solar, and biomass. These systems help reduce dependence on conventional fossil fuels and support environmentally friendly energy production.

The adoption of renewable energy systems in urban areas improves energy efficiency and reduces carbon emissions. They play a significant role in developing smart cities by providing decentralized power generation and promoting sustainable urban development.

2.2 SMALL WIND TURBINE TECHNOLOGIES

Small-scale wind turbine technologies are developed to generate electricity from low and moderate wind speeds in residential and urban areas. These turbines are compact, lightweight, and designed for decentralized power generation applications.

Modern small wind turbines use efficient blade designs, permanent magnet generators, and advanced control systems to improve energy conversion efficiency. They provide a reliable renewable energy solution and can be integrated with battery storage and IoT monitoring systems for enhanced performance.

2.3 IoT-BASED ENERGY MONITORING

IoT-based energy monitoring systems use sensors, microcontrollers, and internet connectivity to collect and transmit energy-related data in real time. These systems enable users to monitor parameters such as voltage, current, power consumption, and energy generation from remote locations.

The integration of IoT improves system efficiency, data accuracy, and energy management. In renewable energy applications, IoT platforms such as Blynk provide real-time visualization, remote access, and performance analysis, helping users optimize energy utilization and system operation.

2.4 ESP32 APPLICATIONS IN RENEWABLE ENERGY

The ESP32 is a powerful microcontroller widely used in renewable energy systems due to its built-in Wi-Fi and Bluetooth capabilities. It enables real-time monitoring and control of renewable energy sources such as wind turbines, solar panels, and battery storage systems.

In renewable energy applications, the ESP32 collects data from sensors measuring voltage, current, power, and environmental conditions. The processed data can be displayed locally and transmitted to IoT platforms for remote monitoring, improving system efficiency and energy management.

2.5 ENERGY STORAGE SYSTEMS

Energy storage systems are essential components of renewable energy projects because energy generation from sources such as wind and solar is often intermittent. These systems store excess electrical energy and supply power when generation is low or unavailable.

Rechargeable batteries, such as lithium-ion batteries, are commonly used for energy storage due to their high efficiency and long life. In mini wind turbine

systems, energy storage ensures a stable power supply and improves the overall reliability and performance of the renewable energy system.

2.6 RESEARCH GAP

Several existing studies have focused on large-scale wind energy systems and general renewable energy monitoring. However, many of these systems are expensive, complex, and not suitable for small-scale urban applications where space and installation constraints exist.

There is a need for a low-cost and compact wind energy generation system integrated with real-time IoT monitoring. The proposed project addresses this gap by combining a mini wind turbine, ESP32 microcontroller, energy storage system, and Blynk-based monitoring platform to provide an efficient solution for urban renewable energy generation.

CHAPTER 3

PROBLEM IDENTIFICATION

3.1 PROBLEM IDENTIFIED

The increasing demand for electricity in urban areas and the depletion of conventional energy resources have created a need for sustainable energy solutions. Large wind turbines are not suitable for cities due to space constraints and high installation costs. In addition, many small-scale renewable energy systems lack efficient monitoring and control mechanisms. Therefore, there is a need for a compact mini wind turbine integrated with IoT technology to generate clean energy and provide real-time performance monitoring in urban environments.

3.2 LIMITATIONS OF CONVENTIONAL POWER SOURCES

Conventional power sources such as coal, oil, and natural gas are non-renewable and are being depleted due to continuous usage. Their availability is limited, making them unsuitable for meeting long-term energy demands.

The use of fossil fuels produces harmful emissions that contribute to air pollution and climate change. In addition, rising fuel costs and increasing energy demand create economic and environmental challenges for sustainable development.

3.3 NEED FOR SMART RENEWABLE SYSTEMS

The increasing demand for clean energy requires renewable energy systems that are efficient, reliable, and easy to monitor. Traditional renewable energy systems often lack real-time monitoring and performance analysis capabilities.

Smart renewable systems integrate sensors, microcontrollers, and IoT technology to monitor energy generation and system performance continuously. These systems improve energy management, enhance efficiency, and enable users to access real-time data remotely for better decision-making.

3.4 PROPOSED SOLUTION

The proposed solution is to develop a Mini Wind Turbine for Urban Energy Generation that can convert wind energy into electrical energy and store it in a rechargeable battery. The system is designed to operate efficiently in urban environments where wind speeds are relatively low.

An ESP32 microcontroller is used to monitor wind speed, voltage, current, and power generation in real time. The collected data is displayed on an OLED screen and transmitted to the Blynk IoT platform, enabling remote monitoring and efficient energy management.

3.5 OBJECTIVES

The main objective of the proposed system is to generate electrical energy from urban wind resources using a mini wind turbine and store the generated power in a rechargeable battery. The system aims to provide a clean, renewable, and sustainable source of energy for small-scale applications.

Another objective is to implement real-time monitoring of wind speed, voltage, current, and power generation using an ESP32 microcontroller and IoT technology. This enables efficient energy management, remote monitoring, and improved system performance through continuous data analysis.

CHAPTER 4 METHODOLOGY

4.1 PROJECT METHODOLOGY

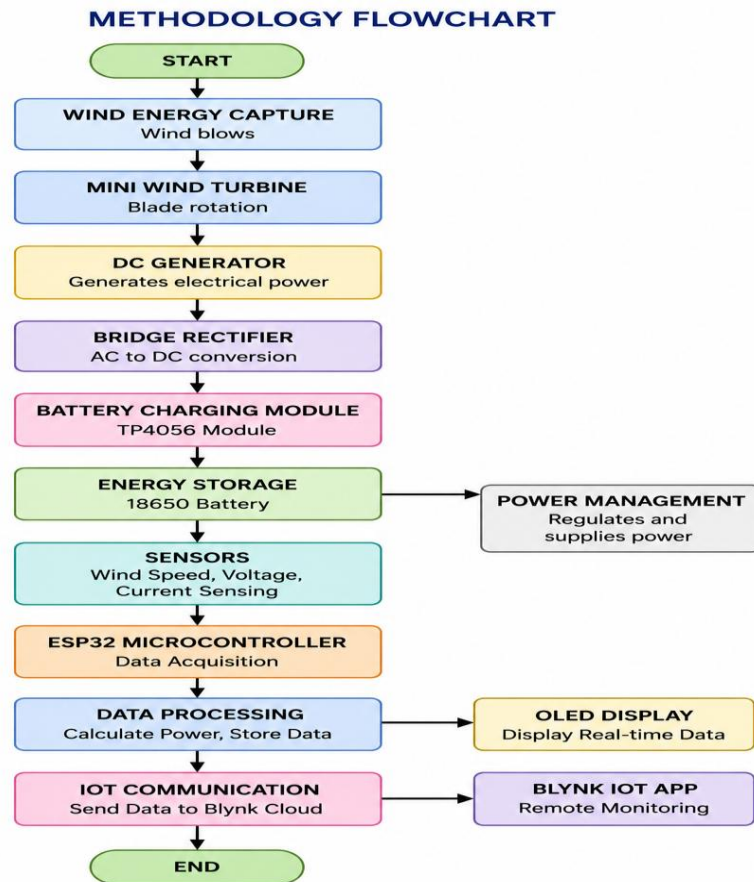


FIG 4.1 METHODOLOGY FLOWCHART

4.2 SYSTEM ARCHITECTURE

The system architecture consists of a mini wind turbine, power conversion circuit, energy storage unit, sensors, and an ESP32 microcontroller. Wind energy is converted into electrical energy, regulated, and stored in a rechargeable battery for continuous operation.

The ESP32 collects data from the voltage, current, and wind speed sensors to monitor system performance. The measured parameters are displayed on an OLED screen and transmitted to the Blynk IoT platform for real-time remote monitoring and analysis.

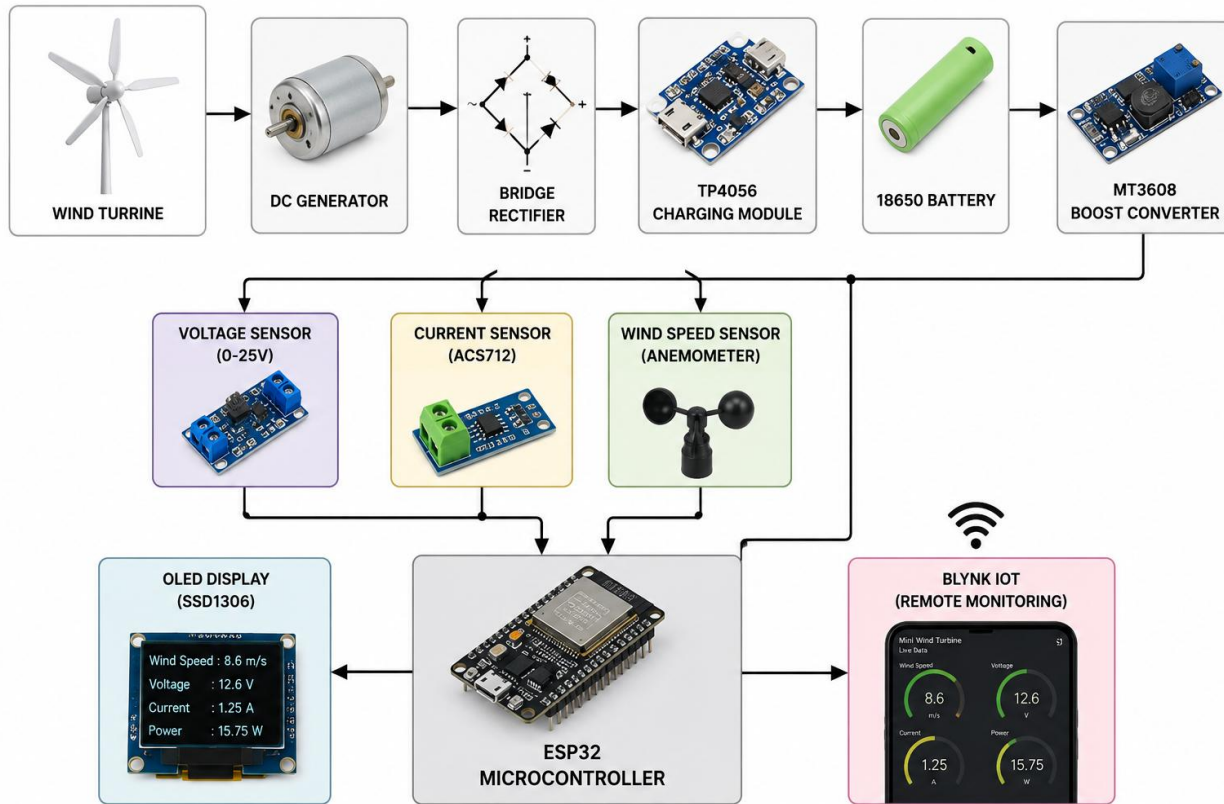


FIG 4.2 SYSTEM BLOCK DIAGRAM

4.3 ENERGY GENERATION PROCESS

The energy generation process begins when wind strikes the blades of the mini wind turbine, causing them to rotate. This rotational motion drives the DC generator, which converts mechanical energy into electrical energy.

The generated electrical power is passed through a bridge rectifier and charging circuit before being stored in a rechargeable battery. The stored energy is then regulated using a boost converter and supplied to the monitoring system and connected loads.

4.4 DATA ACQUISITION PROCESS

The data acquisition process involves collecting real-time information from the voltage sensor, current sensor, and wind speed sensor connected to the system. These sensors continuously measure the operating parameters of the mini wind turbine.

The ESP32 microcontroller receives and processes the sensor data to calculate voltage, current, wind speed, and power generation. The collected information is then displayed on the OLED screen and transmitted to the Blynk IoT platform for remote monitoring and analysis.

4.5 IoT MONITORING METHOD

The IoT monitoring method uses the ESP32 microcontroller to collect data from the voltage, current, and wind speed sensors in real time. The ESP32 processes the sensor readings and connects to the internet through Wi-Fi.

The monitored data is transmitted to the Blynk IoT platform, where users can view system performance remotely using a smartphone or computer. This enables continuous monitoring, data analysis, and efficient management of the wind energy generation system.

CHAPTER 5

COMPONENT SELECTION

5.1 WIND TURBINE BLADES



FIG. 5.1 WIND TURBINE

Wind turbine blades are the primary components responsible for capturing wind energy and converting it into rotational motion. In this project, the blades are fabricated from PVC pipes due to their lightweight structure, low cost, and ease of shaping. The aerodynamic design of the PVC blades improves wind energy capture and enhances the overall efficiency of the mini wind turbine system.

5.2 DC GENERATOR



FIG 5.2 DC GENERATOR

A DC generator is used to convert the mechanical energy produced by the rotating wind turbine blades into electrical energy. When the turbine blades rotate due to wind flow, the generator produces a DC voltage output. The generated electrical energy is then supplied to the charging and storage system for further utilization in the mini wind turbine setup.

5.3 ESP32 CONTROLLER

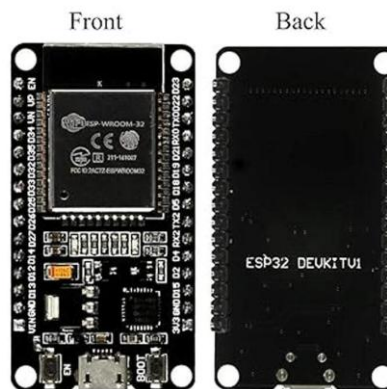
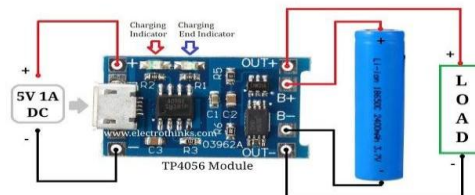


FIG 5.3 ESP32 BOARD

The ESP32 Development Board is the main controller used in the mini wind turbine system for data acquisition, processing, and IoT communication.

5.4 TP4056 CHARGING MODULE

TP4056 Charging Module



with Protection

FIG 5.4 TP4056 MODULE

The TP4056 charging module is used to safely charge the 18650 lithium-ion battery using the electrical energy generated by the mini wind turbine. It provides overcharge, over-discharge, and short-circuit protection to ensure safe battery operation. The module efficiently manages the charging process and helps improve the reliability and lifespan of the energy storage system.

5.5 18650 BATTERY



FIG 5.5 BATTERY

The 18650 lithium-ion battery is used as the energy storage device in the mini wind turbine system. It stores the electrical energy generated by the wind turbine and supplies power to the ESP32, sensors, and other connected components when required. Due to its high energy density, rechargeable nature, and long lifespan, the 18650 battery is widely used in renewable energy and portable electronic applications.

5.6 MT3608 BOOST CONVERTER



FIG 5.6 MT3608 CONVERTER

The MT3608 Boost Converter is a DC-DC step-up voltage regulator used to increase the battery output voltage to the required operating level. In the mini wind turbine system, it boosts the voltage from the 18650 lithium-ion battery to provide a stable power supply for the ESP32, sensors, and display modules. The converter offers high efficiency, compact size, and reliable voltage regulation, making it suitable for renewable energy applications.

5.7 VOLTAGE SENSOR



FIG 5.7 VOLTAGE SENSOR

The Voltage Sensor Module is used to measure the output voltage generated by the mini wind turbine and battery system. It converts higher voltages into a safe range that can be read by the ESP32 analog input pins. The measured voltage data is processed by the ESP32 and displayed on the OLED screen and Blynk IoT platform for real-time monitoring.

5.8 ACS712 CURRENT SENSOR

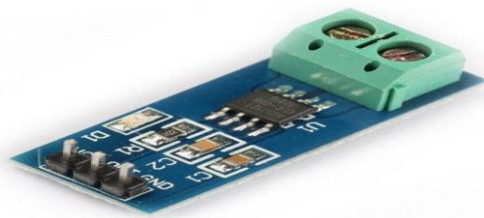


FIG 5.8 CURRENT SENSOR

The ACS712 Current Sensor is used to measure the current generated by the mini wind turbine and supplied to the load or battery. It operates based on the Hall-effect principle and provides an analog output proportional to the current flowing through the circuit. The sensor data is read by the ESP32 and displayed on the OLED screen and Blynk IoT platform for real-time current monitoring.

5.9 WIND SPEED SENSOR



FIG 5.9 WIND SPEED SENSOR

The Wind Speed Sensor is used to measure the velocity of wind available for power generation in the mini wind turbine system. It continuously monitors wind speed and sends the measured data to the ESP32 microcontroller for processing and analysis. The recorded wind speed values are displayed on the OLED screen and uploaded to the Blynk IoT platform for real-time monitoring of turbine performance.

5.10 OLED DISPLAY



FIG 5.10 OLED DISPLAY

The OLED (Organic Light Emitting Diode) Display is used to show real-time system parameters such as voltage, current, power generation, and wind speed. It receives processed data from the ESP32 microcontroller and provides a clear visual representation of the system status. The OLED display is compact, energy-efficient, and offers high contrast, making it suitable for renewable energy monitoring applications.

5.11 LED INDICATOR

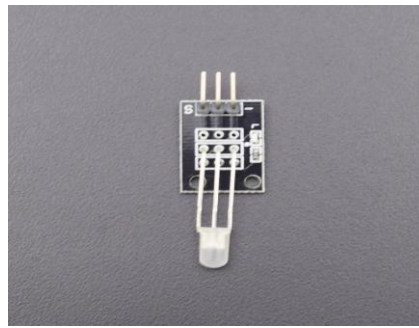


FIG 5.11 LED INDICATOR

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The LED Indicator is used to provide a visual indication of the operational status of the mini wind turbine system. It can indicate power availability, battery charging status, or system operation through simple light signals. The LED consumes very low power and helps users quickly identify the system condition without accessing the display or IoT platform.

CHAPTER 6

SYSTEM OPERATION

6.1 WORKING PRINCIPLE

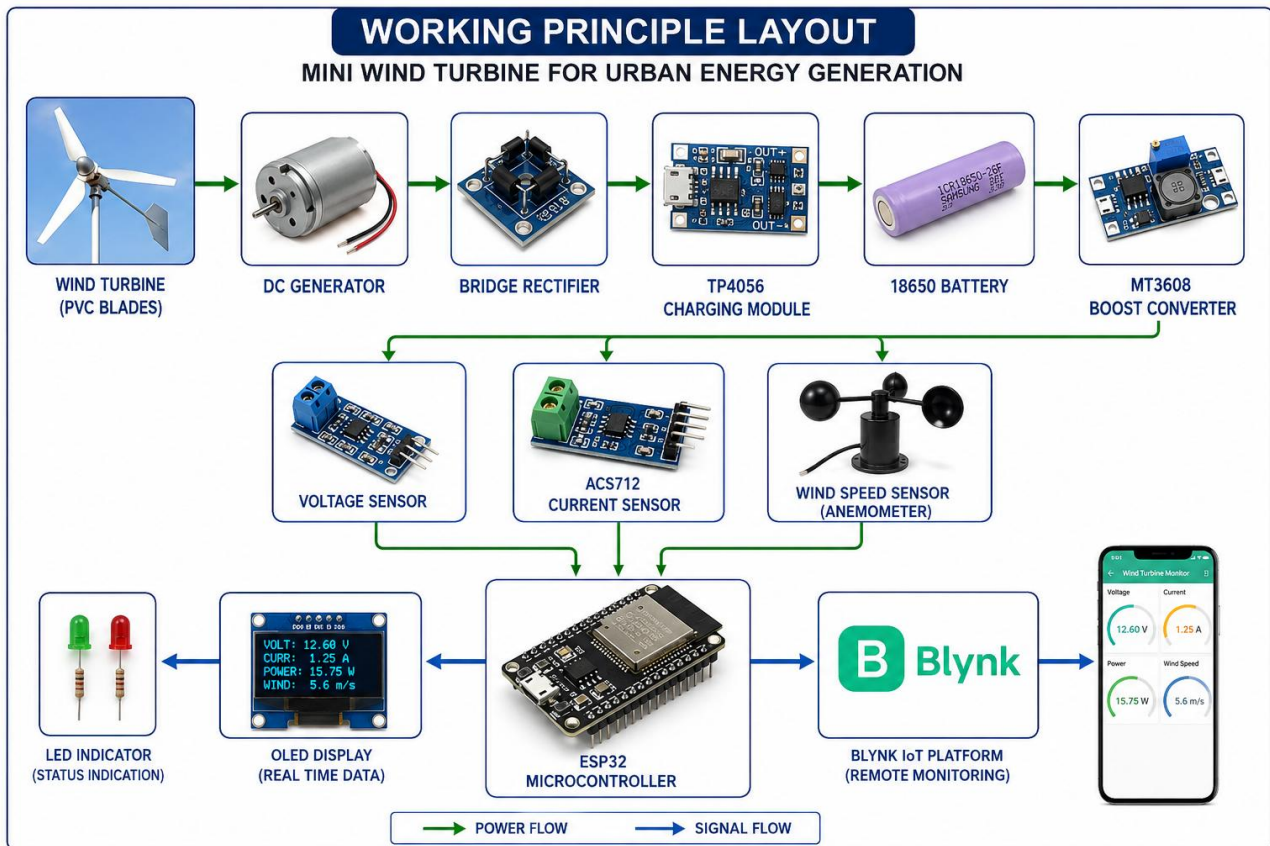


FIG 6.1 WORKING LAYOUT

The working of the mini wind turbine system begins when wind strikes the PVC turbine blades, causing them to rotate. The rotational motion drives the DC generator, which converts mechanical energy into electrical energy. The generated power is passed through a bridge rectifier and TP4056 charging module before being stored in the 18650 lithium-ion battery for later use.

The stored energy is regulated by the MT3608 boost converter and supplied to the ESP32-based monitoring system. The voltage sensor, ACS712 current sensor, and

wind speed sensor continuously measure system parameters and send the data to the ESP32. The processed information is displayed on the OLED screen and transmitted to the Blynk IoT platform, enabling real-time monitoring of wind speed, voltage, current, and power generation.

6.2 POWER GENERATION

The power generation process starts when wind flows over the turbine blades, causing them to rotate. The rotational energy of the blades is transferred to the DC generator, which converts the mechanical energy into electrical energy. The amount of power generated depends on the wind speed and the efficiency of the turbine system.

The electrical energy produced by the generator is converted into a stable DC output using a bridge rectifier and then stored in the 18650 lithium-ion battery through the TP4056 charging module. The stored energy is regulated by the MT3608 boost converter and used to power the monitoring system and connected loads.

6.3 BATTERY CHARGING

The battery charging process begins when the DC generator produces electrical energy from the rotation of the wind turbine blades. The generated power is passed through a bridge rectifier to ensure a stable DC output and then supplied to the TP4056 charging module. The TP4056 module controls the charging current and voltage to safely charge the 18650 lithium-ion battery.

The charged battery stores the generated energy for future use and provides a continuous power supply even when wind speed is low or unavailable. The charging status can be monitored through the voltage sensor and ESP32 controller, while the collected data is displayed on the OLED screen and transmitted to the Blynk IoT platform for real-time monitoring.

6.4 VOLTAGE MONITORING

The voltage monitoring system uses a voltage sensor module to continuously measure the output voltage generated by the wind turbine and the battery. The sensor converts the voltage into a safe analog signal that can be read by the ESP32 microcontroller. This helps in tracking the performance and stability of the power generation system.

The ESP32 processes the voltage readings and displays the measured values on the OLED screen for local monitoring. The voltage data is also transmitted to the Blynk IoT platform, allowing users to monitor the system remotely and ensure efficient battery charging and power management.

6.5 CURRENT MONITORING

The current monitoring system uses the ACS712 current sensor to measure the current generated by the wind turbine and supplied to the battery or load. The sensor operates on the Hall-effect principle and provides an analog output proportional to the current flowing through the circuit. This enables accurate measurement of the system's electrical performance.

The ESP32 microcontroller reads the current sensor data and processes it for monitoring and analysis. The measured current values are displayed on the OLED screen and transmitted to the Blynk IoT platform, allowing users to track power generation and energy consumption in real time.

6.6 WIND SPEED MONITORING

The wind speed monitoring system uses an anemometer sensor to measure the speed of the wind available for power generation. The sensor continuously detects wind velocity and sends the measured data to the ESP32 microcontroller for

processing. Wind speed is an important parameter that directly affects the performance and power output of the mini wind turbine.

The ESP32 calculates and displays the wind speed values on the OLED screen for local monitoring. The measured data is also transmitted to the Blynk IoT platform, enabling users to remotely monitor wind conditions and analyze the relationship between wind speed and power generation in real time.

6.7 OLED DISPLAY OPERATION

The OLED display is used to present real-time information about the performance of the mini wind turbine system. It receives processed data from the ESP32 microcontroller and displays important parameters such as voltage, current, wind speed, and power generation. The display provides a clear and user-friendly interface for local monitoring.

The OLED screen continuously updates the measured values, allowing users to observe changes in system performance instantly. Its low power consumption, high contrast ratio, and compact size make it an ideal display device for renewable energy monitoring applications and portable embedded systems.

6.8 BLYNK IoT MONITORING

The Blynk IoT platform is used to remotely monitor the performance of the mini wind turbine system through a smartphone or computer. The ESP32 microcontroller collects data from the voltage sensor, current sensor, and wind speed sensor and transmits it to the Blynk cloud using a Wi-Fi connection. This enables real-time access to system parameters from any location.

The Blynk dashboard displays important information such as voltage, current, wind speed, power generation, and battery status in graphical and numerical formats.

Users can continuously monitor the system performance, analyze energy generation trends, and ensure efficient operation of the renewable energy system through the IoT platform.

CHAPTER 7

DESIGN CALCULATION

7.1 WIND POWER CALCULATION

The power available in the wind can be calculated using the standard wind power equation. This equation depends on air density, swept area of the turbine blades, and wind velocity. The wind power increases significantly with an increase in wind speed.

Wind Power Equation

$$P = \frac{1}{2} \rho A V^3$$

Where:

- P = Wind Power (W)
- ρ = Air Density (1.225 kg/m³)
- A = Swept Area of Turbine (m²)
- V = Wind Speed (m/s)

Sample Calculation

Given:

- Air Density (ρ) = 1.225 kg/m³
- Blade Radius (r) = 0.15 m
- Wind Speed (V) = 5 m/s

Swept Area,

$$A = \pi r^2$$

$$A = 3.14 \times (0.15)^2$$

$$A = 0.07065 \text{ m}^2$$

Wind Power,

$$P = \frac{1}{2} \times 1.225 \times 0.07065 \times (5)^3$$

$$P = 0.6125 \times 0.07065 \times 125$$

$$P = 5.41 \text{ W}$$

7.2 TURBINE OUTPUT CALCULATION

The actual output power of the wind turbine depends on the turbine efficiency and the available wind power. Since all the wind energy cannot be converted into useful electrical energy, the turbine output is calculated by multiplying the wind power by the turbine efficiency factor.

Turbine Output Equation

$$P_{out} = P_{wind} \times \eta$$

Where:

- P_{out} = Turbine Output Power (W)
- P_{wind} = Available Wind Power (W)
- η = Turbine Efficiency

Sample Calculation

Given:

- Wind Power (P_{wind}) = 5.41 W
- Turbine Efficiency (η) = 35% = 0.35

$$P_{out} = 5.41 \times 0.35$$

$$P_{out} = 1.89 \text{ W}$$

7.3 BATTERY CHARGING CALCULATION

The battery charging time depends on the battery capacity and the charging current supplied by the wind turbine system. The charging time can be estimated by dividing the battery capacity by the charging current. This calculation helps determine the duration

required to fully charge the battery.

Battery Charging Equation

$$T = \frac{C}{I}$$

Where:

- T = Charging Time (hours)
- C = Battery Capacity (mAh)
- I = Charging Current (mA)

Sample Calculation

Given:

- Battery Capacity = 2200 mAh
- Charging Current = 500 mA

$$T = \frac{2200}{500}$$

$$T = 4.4 \text{ hours}$$

7.4 POWER CONSUMPTION ANALYSIS

Power consumption analysis is used to determine the amount of electrical power required by the components in the mini wind turbine monitoring system. The total power consumption is calculated by summing the power used by the ESP32, OLED display, sensors, and other electronic components. This analysis helps evaluate the energy efficiency of the system.

Power Consumption Equation

$$P = V \times I$$

Where:

- P = Power Consumption (W)
- V = Supply Voltage (V)

- I = Current Consumption (A)

Sample Calculation

Given:

Component	Voltage (V)	Current (A)
ESP32	5	0.08
OLED Display	5	0.02
Sensors	5	0.03

Total Current:

$$I_{total} = 0.08 + 0.02 + 0.03$$

$$I_{total} = 0.13A$$

Total Power Consumption:

$$P = 5 \times 0.13$$

$$P = 0.65W$$

7.5 EFFICIENCY CALCULATION

Efficiency calculation is used to determine how effectively the mini wind turbine converts the available wind energy into useful electrical energy. It is expressed as the ratio of actual electrical output power to the total wind power available at the turbine. A higher efficiency indicates better performance of the wind energy conversion system.

Efficiency Equation

$$\eta = \frac{P_{out}}{P_{wind}} \times 100$$

Where:

- η = Efficiency (%)
- P_{out} = Actual Output Power (W)
- P_{wind} = Available Wind Power (W)

Sample Calculation

Given:

- Available Wind Power = 5.41 W
- Actual Output Power = 1.89 W

$$\eta = \frac{1.89}{5.41} \times 100$$

$$\eta = 34.93\%$$

$$\eta \approx 35\%$$

CHAPTER 8

RESULTS AND DISCUSSION

8.1 EXPERIMENTAL SETUP

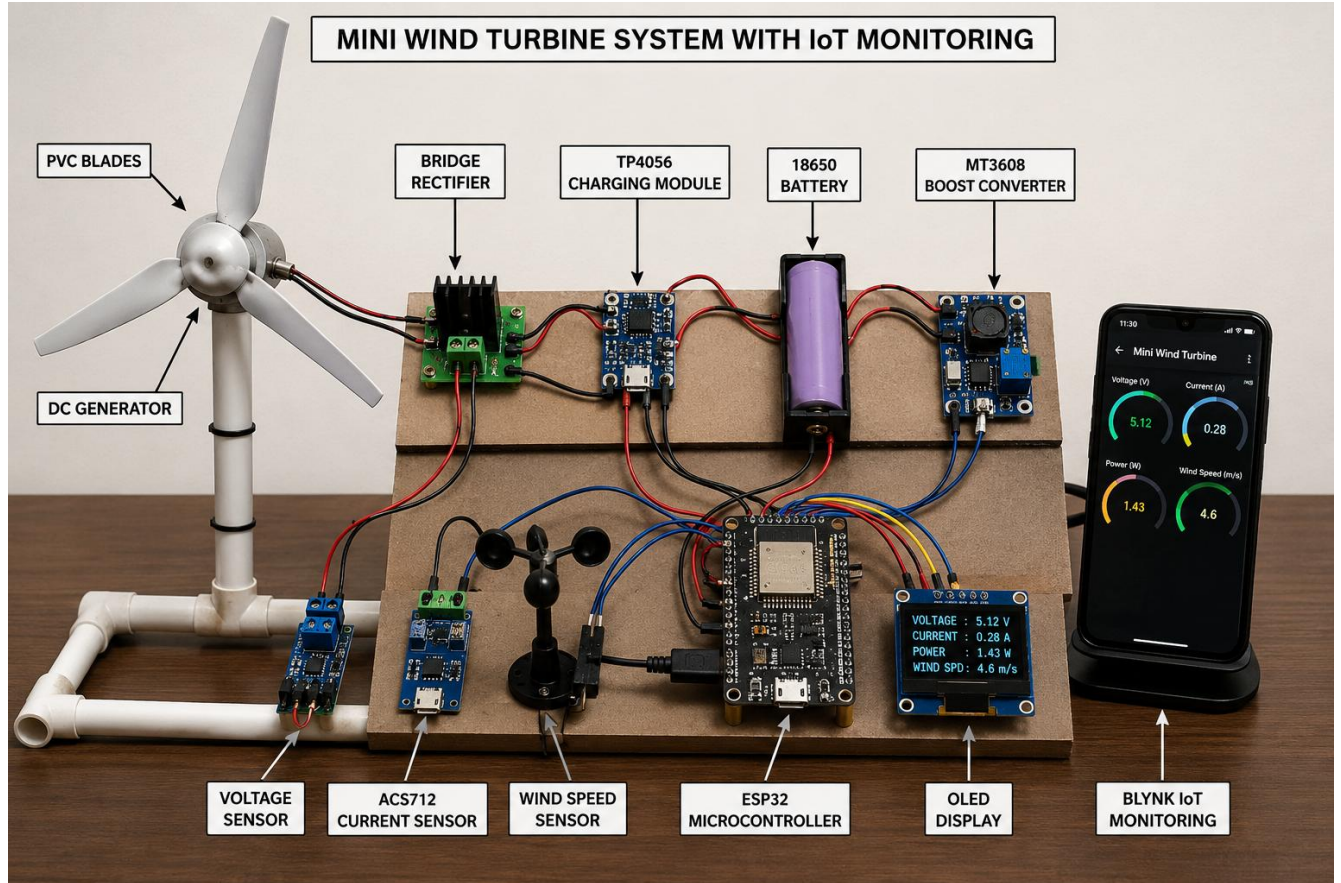


FIG 8.1 PROTOTYPE SETUP

TABULATION

Trial No	Wind Speed (m/s)	Voltage (V)	Current (A)	Power (W)
1	2.0	1.8	0.10	0.18
2	3.0	2.6	0.15	0.39
3	4.0	3.5	0.20	0.70
4	5.0	4.5	0.25	1.13
5	6.0	5.6	0.30	1.68

TABLE 8.1 EXPERIMENTAL RESULTS

8.2 VOLTAGE OUTPUT RESULTS

The voltage output of the mini wind turbine was analyzed to evaluate the electrical energy generated under different wind conditions. As the wind speed increased, the rotational speed of the turbine blades also increased, causing the DC generator to produce a higher voltage output. This demonstrates the direct relationship between wind velocity and electrical power generation.

The voltage sensor module successfully measured the generated voltage and transmitted the data to the ESP32 controller for processing. The measured values were displayed on the OLED screen and uploaded to the Blynk IoT platform for real-time monitoring. The results indicate that the proposed system can effectively generate and monitor electrical energy, making it suitable for small-scale urban renewable energy applications.

8.3 CURRENT OUTPUT RESULTS

The current output of the mini wind turbine system was analyzed to evaluate the amount of electrical current generated under varying wind conditions. As the wind speed increased, the turbine blades rotated at a higher speed, resulting in greater electrical energy production and an increase in output current. This indicates that the current generated by the system is directly influenced by the available wind energy.

The ACS712 current sensor successfully measured the generated current and transmitted the readings to the ESP32 controller for processing and monitoring. The measured current values were displayed on the OLED screen and uploaded to the Blynk IoT platform for real-time observation. The results demonstrate that the proposed system can effectively generate and monitor electrical current, making it suitable for small-scale urban wind energy applications.

8.4 WIND SPEED RESULTS

The wind speed results were analyzed to evaluate the performance of the mini

wind turbine under different operating conditions. The wind speed sensor continuously measured the velocity of the incoming wind and transmitted the data to the ESP32 controller. The recorded wind speed values played a crucial role in determining the amount of electrical energy generated by the turbine.

The experimental results showed that an increase in wind speed led to a corresponding increase in turbine rotational speed, voltage output, current output, and power generation. The measured wind speed data was displayed on the OLED screen and uploaded to the Blynk IoT platform for real-time monitoring and analysis. These results confirm that wind speed is a key factor influencing the overall performance and efficiency of the mini wind turbine system.

8.5 POWER GENERATION ANALYSIS

The power generation analysis was conducted to evaluate the electrical power produced by the mini wind turbine under different wind conditions. The generated power was calculated using the measured voltage and current values obtained from the voltage sensor and ACS712 current sensor. The results showed that power generation increased with an increase in wind speed, demonstrating the effectiveness of the wind energy conversion process.

The ESP32 controller continuously monitored and processed the power generation data, displaying the results on the OLED screen and transmitting them to the Blynk IoT platform for remote monitoring. The analysis confirmed that the mini wind turbine is capable of generating useful electrical energy for small-scale applications. The obtained results indicate that the proposed system can serve as a reliable and sustainable solution for urban renewable energy generation.

8.6 IoT MONITORING RESULTS

The IoT monitoring system was successfully implemented using the ESP32 microcontroller and the Blynk IoT platform. The system continuously collected data

from the voltage sensor, ACS712 current sensor, and wind speed sensor and transmitted the information to the cloud through a Wi-Fi connection. This enabled real-time monitoring of important system parameters from any location using a smartphone or computer.

The results demonstrated that the Blynk platform effectively displayed voltage, current, wind speed, power generation, and battery status in both numerical and graphical formats. The data was updated in real time, allowing users to analyze system performance and energy generation trends efficiently. The successful operation of the IoT monitoring system confirms the effectiveness of integrating renewable energy systems with smart monitoring technology for improved energy management and remote accessibility.

CHAPTER 9

CONCLUSION

The mini wind turbine for urban energy generation was successfully designed and developed to harness renewable wind energy for small-scale power generation. The system effectively converted wind energy into electrical energy using a DC generator and stored the generated power in a rechargeable battery. The project demonstrated the feasibility of utilizing wind resources in urban environments for sustainable energy production.

The hardware components, including the ESP32 controller, voltage sensor, ACS712 current sensor, wind speed sensor, OLED display, and Blynk IoT platform, worked efficiently as an integrated monitoring system. Real-time measurement and display of voltage, current, wind speed, and power generation were achieved successfully. The monitoring system provided accurate and reliable data for evaluating the performance of the wind turbine.

The experimental results showed that the output voltage, current, and power generation increased with increasing wind speed. The battery charging circuit effectively stored the generated energy and ensured continuous power availability during low wind conditions. The obtained results confirmed that the proposed system can generate useful electrical energy and support low-power electronic applications.

The integration of IoT technology enhanced the functionality of the system by enabling remote monitoring and data visualization through the Blynk platform. The project highlights the potential of combining renewable energy with smart technologies for efficient energy management. Overall, the developed mini wind turbine system offers a cost-effective, eco-friendly, and sustainable solution for urban renewable energy generation.

CHAPTER 10

FUTURE SCOPE

The proposed mini wind turbine system can be further enhanced by using larger and more efficient turbine blades to increase power generation capacity. Advanced generator designs and improved aerodynamic blade structures can help maximize energy conversion efficiency. This would make the system suitable for powering a wider range of household and commercial applications.

Future development can include the integration of solar panels to create a hybrid renewable energy system. Combining wind and solar energy sources can ensure continuous power generation under varying weather conditions. Such a hybrid approach can improve system reliability and overall energy availability.

The IoT monitoring platform can be upgraded with cloud data analytics, machine learning, and predictive maintenance features. These technologies can analyze historical performance data, predict energy generation trends, and identify potential system faults before failure occurs. This will enhance system efficiency and reduce maintenance requirements.

The project can also be scaled for deployment in residential buildings, educational institutions, and smart city applications. Advanced battery storage systems and smart energy management techniques can be incorporated for efficient power utilization. The future implementation of such systems can contribute significantly to sustainable urban energy.

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PROTOTYPE PHOTOGRAPH

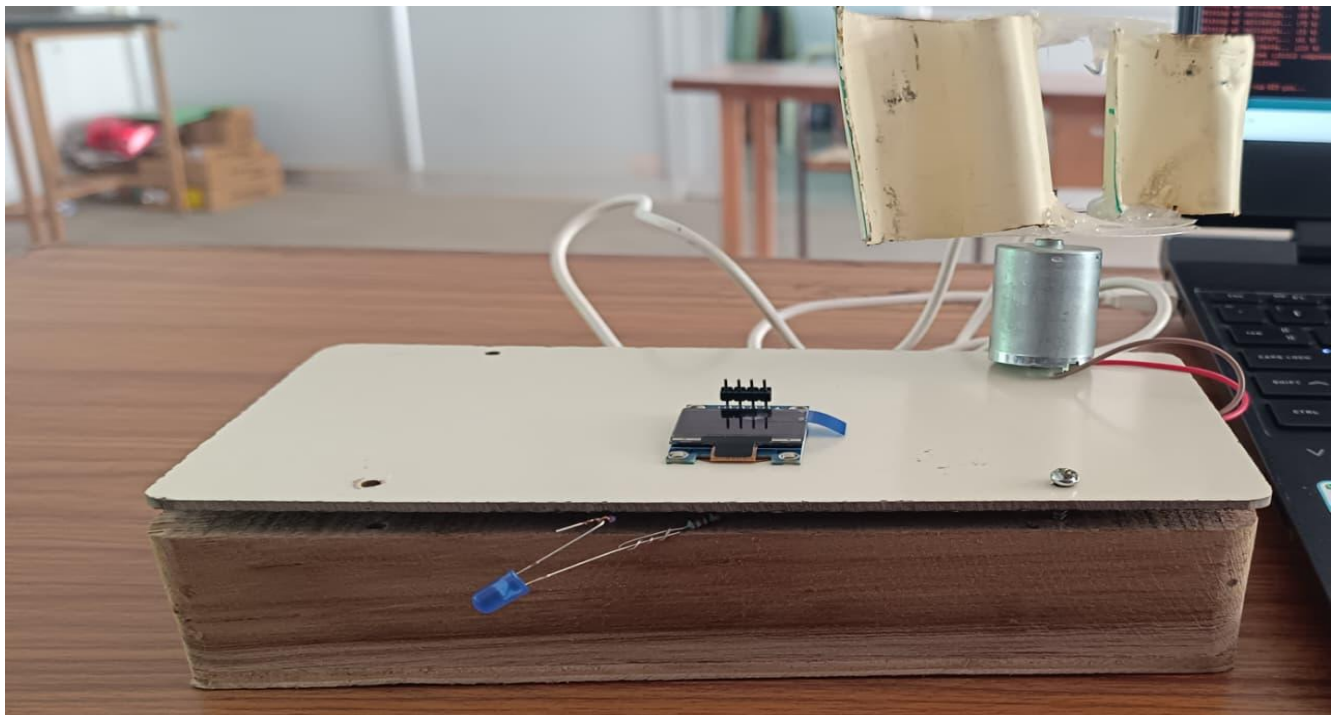
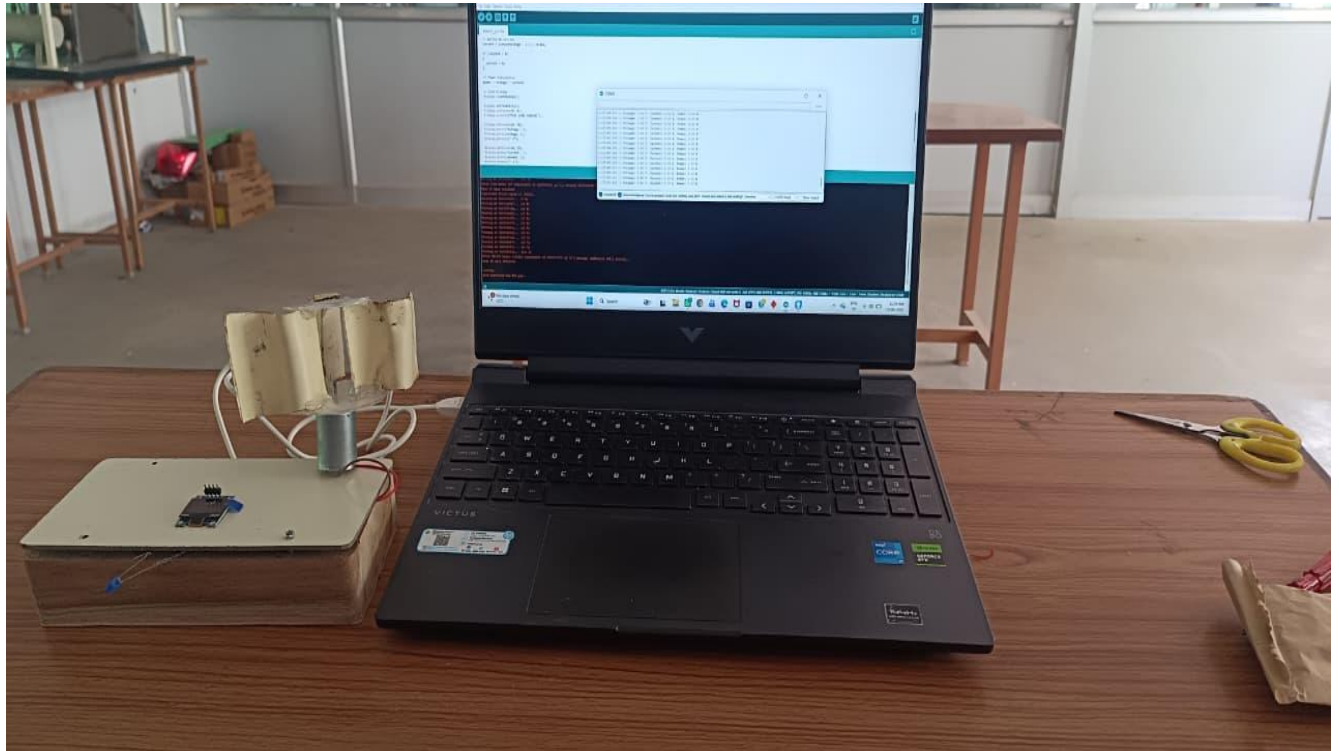


FIG. 11.1 Prototype Photograph